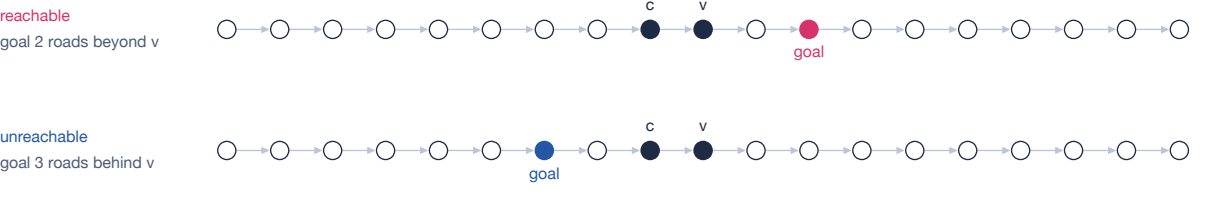
The lower bound for one layer (L = 1): two paths of 19 towns that one layer cannot tell apart



Roads all point right. From v the top goal is reached in 2 roads; the bottom goal lies behind v and is never reached.
After one round of folklore 2-WL the pair (c, v) has the same colour in both, so no one-layer network of the class tells them apart.